Online Appendix

Manuscript: *Visibility Bias in Chinese Municipal Government Work Reports: Measurement and Behavioral Validation*

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Appendix A: The Full Cadre-Attention Model

The main text (§2) gives an intuitive summary of the cadre-attention model. The full derivation follows.

A.1 Setup

A local official in city i allocates a municipal capital budget B_i between visible capital V_i and functional capital F_i subject to $V_i + F_i = B_i$. Social welfare is Cobb-Douglas:

$$U^S(V_i, F_i) = \alpha \ln V_i + \beta \ln F_i, \alpha + \beta = 1.$$

The supervisor observes both categories with Gaussian noise with variances $\omega_V^2 < \omega_F^2$ (assumption A1, observability asymmetry). The Bayesian posterior on the official's performance is the precision-weighted sum:

$$\pi_i = \rho V_i + (1-\rho) F_i, \rho = \frac{\omega_V^{-2}}{\omega_V^{-2} + \omega_F^{-2}} \in (1/2, 1).$$

A.2 Official's objective and FOC

The official maximizes a convex combination of social welfare and career value:

$$U^O = \lambda_i \left[\alpha \ln(a_i B_i) + \beta \ln((1-a_i) B_i)\right] + (1-\lambda_i) B_i \left[\rho a_i + (1-\rho)(1-a_i)\right]$$

where $a_i = V_i / B_i \in (0,1)$ and $\lambda_i \in [0,1]$ is the social-motivation weight. The first-order condition yields the cadre-attention equation:

$$\frac{\alpha}{a_i^*} - \frac{\beta}{1-a_i^*} = -\frac{1-\lambda_i}{\lambda_i} \phi B_i, \phi \equiv 2\rho - 1 > 0.$$

A.3 Closed-form and welfare loss

A Taylor expansion around $\delta_i \equiv (1-\lambda_i)/\lambda_i \cdot \phi B_i = 0$ gives:

$$a_i^* \approx a_i^{SO} + a_i^{SO}(1 - a_i^{SO}) \delta_i.$$

The welfare loss expressed as a fraction of first-best utility is:

$$W_i = \frac{1}{2} \cdot \frac{(a_i^* - a_i^{SO})^2}{a_i^{SO}(1 - a_i^{SO})}.$$

This quadratic form allows calibration from observed a_i^* together with a standard social-optimum benchmark a_i^{SO}.

Appendix B: Specification Curve (P1 Robustness — demoted)

This appendix supports the cross-sectional VAI→CIR association (§4.1), a modest external check (E-D, §3.2.4). It is not part of the main evidentiary spine, which rests on the behavioral criterion-validity co-movement of §3.2.7.

B.1 24-permutation specification curve

Following Simonsohn et al. [17], we ran a pre-registered specification curve combining: - Lexicon thresholds: denominator requirement {≥3, ≥5, ≥10 total hits} - Control-set subsets: {all controls, drop ln_pop, drop ind2_share, no controls} - Panel windows: {2005-2015 full, 2008-2015 post-GFC, 2012-2015 post-Xi} - FE structure: {city + year FE, city FE only}

Total: 3 × 4 × 3 × 2 / redundant = 24 non-redundant permutations.

Results: Median β = +0.107. All 24 coefficients positive. 22/24 significant at the 5% level; 2 marginally significant (p = 0.06–0.10). Full specification curve table in `03-analysis/phase-A/specification_curve_v2.py`.

Appendix C: Further Robustness and Sensitivity

C.1 Dictionary details

Full lists of V_ORIG (42 terms), F_ORIG (38 terms), V_EXT (+36 new terms, 78 total), F_EXT (+32 new terms, 70 total) are in `03-analysis/phase-E/construct_validity.py`. Coder-agreement statistics from the inter-coder validation protocol (Cohen's κ): - Visible-term marking: κ = 0.87 - Functional-term marking: κ = 0.84 - Overall orientation (1–5 Likert): ICC = 0.79

C.2 Horse races against alternative explanations (main text §4.1)

Alternative	Specification	β(VAI)	p
Baseline P1	City FE + year FE + 3 controls	+0.113	0.002
+ GDP growth control	Added annual GDP growth	+0.109	0.003
+ urbanization² control	Added urbanization rate and its square	+0.111	0.002
+ leader-tenure	Added party-	+0.110	0.003

Alternative	Specification	β(VAI)	p
control	secretary + mayor tenure		
All four jointly	All three above	+0.108	0.004

C.3 P2 placebo battery (main text §4.5)

Table C.3. Placebo tests for the inspection event study.

Placebo	β(k=0)	p	Interpretation
Close pre-era shift (fake 2011/2012)	+0.001	0.98	Clean null — supports narrow-window TWFE
Far-era shift +4 yr (fake 2017/2018)	+0.023	0.0003	Significant — flags secular-trend contamination
Far-era shift +6 yr (fake 2019/2020)	−0.025	0.001	Significant — flags secular-trend contamination
Random-assignment (500 iters)	Observed −0.065 vs placebo [−0.043, +0.076]	1.00	Underpowered (only 2 cohorts)

C.4 Evidence for observability asymmetry (A1)

Three independent proxies for ω_V / ω_F:

1. **Inspection frequency**: Municipal Transparency Report audits per year of visible vs functional categories. Ratio 3.2:1 in favor of visible.
2. **Media reporting intensity**: Xinhua local news mentions per month. Ratio 2.8:1 in favor of visible.
3. **Accident-record visibility**: Ministry of Emergency Management major-accident reports by category. Only visible-category accidents (street collapse, façade fall) receive consistent national news coverage. Functional failures (pipe rupture, flooding) are usually reported only as aggregate statistics.

All three support A1 at $p < 0.05$ (separately and jointly).

C.5 Tenure-cycle dynamics (supplementary evidence)

Event-study around scheduled leadership transition (mayor + party-secretary tenure-end year):

Event time	β(VAI)	p
k = −2 (pre-transition −2 years)	+0.008	0.52
k = −1 (year before	**+0.031**	**0.021**

Event time	β(VAI)	p
transition)		
k = 0 (transition year)	−0.009	0.58
k = +1	+0.004	0.78

Interpretation: VAI spikes in the year before a scheduled leadership transition, when career-concern intensity $(1-\lambda)/\lambda$ is theoretically highest. This is consistent with P2's exogenous-attenuation logic in reverse (career-concern spikes → VAI spikes). We treat this as *supplementary* evidence because the identification relies on transitions being partially exogenous, which is weaker than our pre-registered design. Results replicate under city + year FE and are insensitive to controls.

C.6 Heterogeneity by fiscal capacity

The cadre-attention first-order condition (Online Appendix A.2) predicts that the visibility-bias distortion increases in B_i. We split the sample by above/below-median per-capita fiscal revenue:

Subsample	β(VAI) on CIR	p	N
Above-median fiscal revenue	+0.184	0.001	1,378
Below-median fiscal revenue	+0.068	0.071	1,373
Interaction (above × VAI)	+0.116	**0.024**	2,751

The high-revenue subsample's coefficient is substantially larger and more significant. The interaction is statistically distinguishable (p = 0.024).

C.7 Welfare sensitivity (main text §4.3)

Full sensitivity across $a^{SO} \in [0.40, 0.50]$, utility forms, and shadow-pricing assumptions:

a^{SO}	Utility	W per ¥1000B renewal	Scaled to ¥3.5T/yr × 8% renewal share
0.50	Log	0.06%	¥0.2B/yr
0.45	Log (central)	1.56%	**¥4.4B/yr**
0.40	Log	3.94%	¥11.0B/yr
0.45	CRRA(γ=2)	3.12%	¥8.8B/yr
0.45	BOE (replacement cost)	~20%	¥55B/yr

Defensible range: ¥1–15B/yr under log utility and $a^{SO} \in [0.40, 0.50]$. The ¥55B BOE is a ceiling. (This calibration is demoted; the main text attaches no headline welfare figure. See §4.4.)

C.8 Passage-validation confusion matrix and error taxonomy (main text §3.2.0)

Human gold standard: 120 sentences, four-class coding (visible / functional / mixed / irrelevant). Source: `03-analysis/phase-K/` (`anchor_human_labels.csv`, `score_anchor.py`).

Table C.8a. Concrete (salience-based) dictionary vs human coding, n = 120. Rows = human label, columns = dictionary prediction.

human ↓ / dict →	visible	functional	mixed	irrelevant
visible	15	1	1	2
functional	0	18	1	1
mixed	9	7	24	0
irrelevant	6	4	4	27

Per class: visible P = 0.500, R = 0.789, F1 = 0.61; functional P = 0.600, R = 0.900; mixed P = 0.800, R = 0.600; irrelevant P = 0.900, R = 0.659. Overall four-class accuracy = 0.74.

Table C.8b. Visible-class false-positive taxonomy (naive dictionary). Of 84 naive visible-class false positives (dictionary = visible, human/ensemble = not infrastructure), the drivers are:

term	false positives	typical non-infrastructure use
示范 *shìfàn* (model/demonstration)	60–61	示范区, 示范项目, 改革示范
文明城市 *wénmíng chéngshì* (civilized city)	7	award/campaign title
展示 *zhǎnshì* (display/showcase)	6	“display achievements”
美丽 *měilì* (beautiful)	5	美丽乡村 slogan
形象 *xíngxiàng* (image)	4	“government image”

示范 alone accounts for ~70% of visible false positives. This single-cause failure motivated the move to a concrete, salience-based lexicon (§3.2.2). Blanket-pruning the 13 most polysemous terms raises visible precision to 0.59 but collapses recall to 0.47, so the fix is surgical.

Human intercoder agreement. The 120-sentence gold standard was independently double-coded by two coders on the identical locked sentence set (`03-analysis/phase-K/second_coder_sheet_LOCKED.csv`, with dictionary and ensemble labels hidden). The **human intercoder agreement is Cohen κ = 0.70** (substantial; n = 120). Agreement is highest on the irrelevant and functional classes. The residual disagreement concentrates on the visible-versus-mixed boundary, consistent with the polysemy difficulty documented in §3.2.0, which is itself part of why high passage precision favors the LLM ensemble. Human↔ensemble agreement is Cohen κ = 0.66.

Figure ESM-6 (procurement, §4.3). `04-figures/fig_bidding_by_city.{pdf,png}` — visible:functional project-count ratio across Zhejiang localities (2.96M procurement records, 2019–2026; v2 domain-verified lexicon). Panel A: distribution over 108 localities (median 1.8; 97 of 108 above 1:1 parity). Panel B: top 15 localities by procurement volume (near-parity county exceptions in red). Source: `03-analysis/phase-L-bidding/` (`classify_bidding.py`, `bidding_lexicon.py` v2, `by_city.csv`).

Table C.8c. LLM-ensemble (Gemini 3.1 Pro + ChatGPT 5.5 + DeepSeek) vs human, n = 499. Majority label; inter-model Fleiss κ = 0.835; overall accuracy 0.842; human ↔ ensemble Cohen κ = 0.66. Source: `passage_coding_sheet_ensemble.validation.json`.

class	precision	recall
visible	0.793	0.96
functional	0.813	0.897
mixed	0.857	0.609
irrelevant	0.913	0.806

Appendix D: Pre-Registration and Deviation Log

(In main text as §6 Appendix D. See Deviations D-B-1 through D-F-1.)

Appendix E: Replication Architecture

Full directory layout (Zenodo record 19569978):

```
visibility-bias-v2/
├── 00-admin/           Research plan, upgrade roadmap, outreach
drafts
├── 03-analysis/        Python analysis by phase:
│                         G/K  passage validation (human gold + 3-
model LLM ensemble)
│                         J    behavioral criterion validity (turnover
co-movement)
│                         L    Zhejiang procurement-frequency check
│                         B/B2/E/E2/F  demoted inspection /
construct / micro tests
├── 04-figures/         PDF and PNG figures for all main-text exhibits
├── 05-manuscript/      Manuscript sections + compiled R2 + submission
package
├── 06-review/          S7 audit reports + phase memos
├── 07-prereg/          OSF pre-registration archive + Zenodo DOI
record
├── 01-literature/      references.bib
└── 02-data/processed/  text panels (naive + concrete lexicons), CCDI
rounds
```

The criterion-validity panel `master_2002_2024.csv` (merged from the absorbed `officials-turnover-cn` materials; holds both the naive `vai_composite` and concrete `wr_visibility` series alongside CIR and turnover variables) and the lexicon-build script `build_workreport_text.py` are included in the Zenodo record. Running instructions in `README.md` at the record root.

Note: the absorbed `officials-turnover-cn` project is not published separately; its retirement-exogenous identification and `wr_visibility` measure are folded into this paper, so there is no double-publication and no citation of unpublished work.